

A foundational introduction to computer networking as part of a comprehensive CCNA 200-301 course, focusing on the primary types of network devices and their specific functions. At its most basic level, a **computer network** is a digital telecommunications network that allows "nodes" (devices) to share resources, such as two PCs connected by a single cable.

End Hosts: Clients and Servers

The most common devices people interact with are **end hosts**, also known as **endpoints**. These are categorized by their role in a specific transaction:

- **Clients:** A client is a device that **accesses a service** provided by a server. Examples include a PC accessing a YouTube video or an iPhone receiving a file via AirDrop.
- **Servers:** A server is a device that **provides functions or services** to clients. While we often imagine servers as powerful hardware in data centers, any device can act as a server if it provides a service, such as a smartphone sending a file to another phone.

Crucially, a single device can act as a client in one situation and a server in another.

Intermediate Devices: Switches and Routers

To connect multiple end hosts together and bridge different networks, specialized hardware is used:

- **Switches:** These devices are used to **forward traffic within a Local Area Network (LAN)**. They typically feature a large number of interfaces (often 24 or more) to aggregate connections from many end hosts like PCs and printers. However, switches cannot connect different LANs or send data over the internet.
- **Routers:** Unlike switches, routers are designed to **provide connectivity between different LANs** and forward traffic over the internet. Routers generally have **fewer network interfaces** than switches because their primary purpose is connecting networks rather than individual end hosts.

Network Security: Firewalls

To protect networks from attackers on the internet, **firewalls** are used to monitor and control traffic based on specific security rules.

- **Network Firewalls:** These are dedicated hardware appliances that filter traffic entering or exiting a network. They can be placed outside the router or inside the internal network.
- **Host-based Firewalls:** These are software applications installed directly on an end host (like Windows Firewall on a PC) to provide an extra line of defense.
- **Next-Generation Firewalls (NGFW):** Modern firewalls that include advanced features like **Intrusion Prevention Systems (IPS)** and more sophisticated filtering capabilities than traditional firewalls.

Summary of Device Functions

Device	Primary Function	Typical Interface Count
Switch	Forwards traffic within a single LAN.	High (e.g., 24+ ports).
Router	Forwards traffic between different networks/internet.	Low.
Firewall	Monitors and filters traffic based on rules.	Variable.
Server	Provides services or resources.	N/A (End Host).
Client	Accesses services or resources.	N/A (End Host).